

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

**Duration : 1 Hour
Total Marks:50**

Annexure A

Analytical Problem Solving

Code of Conduct:

I M.Arshini/192424322 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student :- *M. Arshini*



Student ID: MARSHINI | Register Number: 192424322

1. For a self-driving car operating on city roads, identify its percepts, possible actions, and characterize the operating environment (fully/partially observable, deterministic/stochastic, static/dynamic, discrete/continuous). Justify which agent architecture (simple reflex, model-based, goal-based, or utility-based) is most suitable and why.
2. A cleaning robot is placed in an unknown building with no map available. It can only sense its immediate surroundings. Explain how an Online Search Agent would explore this environment, and discuss the challenges of searching with partial/no prior information compared to offline search.
3. Compare Bidirectional Search with standard Breadth-First Search for finding a path between a start node S and goal node G in a graph with 10 nodes. Illustrate with an example why bidirectional search can significantly reduce time complexity.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

95

Self-Driving car operating on city roads:-

The self-driving car is an intelligent agent that senses its surroundings, processes information, and performs actions automatically, reaching its destination safely. It uses sensors, cameras, radar, LIDAR, GPS, and Artificial Intelligence to make driving decisions.

a) Percepts (Inputs received by the Agent)

Percepts are information collected by the car from its environment-

- * Traffic lights and signals
- * Road signs and speed limits
- * Lane markings
- * GPS location and digital maps
- * Distance from nearby devices
- * Pedestrians and cyclists
- * Obstacles such as parked vehicles, potholes, animals
- * Weather condition (rain, fog, snow)
- * Road conditions

b) possible actions:-

The self-driving car can perform the following actions

- * Accelerate
- * Brake
- * steer left or right
- * change lanes
- * Turn left to right
- * reverse
- * Stop at traffic signals

c) characterization of the environment -

1) observability - partially observable

the environment is partially observable because the car cannot always observe everything due to:-

- * Blind spots
- * Heavy rain
- * Hidden pedestrians
- * Sensor limitations

2) Deterministic or stochastic

- * other drivers behave unpredictably
- * pedestrians may suddenly cross the road
- * weather conditions may change.

cleaning Robot in an unknown Building

A cleaning robot placed in an unknown building has no prior knowledge of the environment. It must explore, learn, clean the building using an online search Agent.

Working of an online-search Agent:

unlike offline search, an online search Agent does not know the complete map

The robot follows these steps

Step 1:-

Start from the initial location

Step 2:-

sense nearby rooms, walls, doors and obstacles

Step 3:-

store newly discovered locations in memory

Step 4:-

move to an unexplored area.

Step 5:- update its internal map after every movement

Step 6:- Avoid already visited locations whenever

Step 7:- Repeat until every accessible room has been explored and cleaned.

2) static or dynamic

The environment is dynamic because

- * vehicles are constantly moving

- * traffic lights change

- * pedestrians keep moving

- * Road conditions may change.

4) Discrete or continuous - continuous

Driving involves continuous values such as

- * speed

- * steering angle

- * distance

- * time

- * Acceleration

d) suitable Agent Architecture

utility-Based Agent

A utility-Based Agent is the most suitable architecture

Reasons:-

- * evaluates multiple possible actions

- * chooses the safest route

- * minimizes travel time

- * Avoid collisions.

Example :-

Suppose the robot starts at room A

$A \rightarrow B \rightarrow C$

$\downarrow$
 $D \rightarrow E$

The robot first explores B, then C, returns to B, explores D, finally reaches E.

As it moves it gradually builds the map.

Online search

No map available

Learns during exploration

Higher exploration cost

more battery consumption

Suitable for unknown environments

Offline search

Complete map available

Plans before movement

Lower exploration cost

Less battery consumption

Suitable for known environments

Advantages:-

- * works without a map
- * learns continuously
- * Adapts to environmental changes
- * Suitable for dynamic environments

Disadvantages:-

- * slower exploration
- * Higher computational cost
- * may not initially find the shortest path.

2) Compare Bidirectional search with Breadth first search
Both Breadth - first search (BFS) and Bidirectional search are uninformed search algorithms used to find the shortest path in an unweighted graph.

Breadth - first search :-

BFS explores the graph level by level starting from the source node until reaches the goal.

Algorithm:-

- * start from node S
- * Insert S into the Queue
- * Remove the first node.

- * visit all unvisited neighbours
- * Repeat until goal is found

time complexity

$$O(b^d)$$

where:-

b = branching factor

d = depth

Bidirectional search:-

Bidirectional search simultaneously perform BFS from:-

* start node (Forward search)

* Goal node (Backward search)

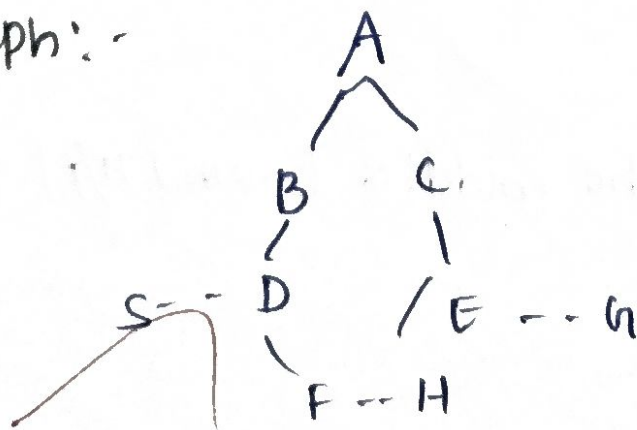
The search stops when both meet.

time complexity

$$O(b^{d/2})$$

EX:- 10 nodes

Graph:-



BFS Traversal:-

S
↓
D
↓
B
↓
F
↓
A
↓
H
↓
C
↓
I
↓
E
↓
G

Bidirectional search

Forward search:-

$S \rightarrow D \rightarrow F$

Backward search

$G \rightarrow E \rightarrow H$

The searches meet near the middle (around H/F) so, far fewer nodes are explored.